

Revised Project Proposal

1 Project Title

Pitch, Register, and Instrument-Family Recognition in NSynth: A Controlled Comparison of Handcrafted Acoustic Features and Pretrained Music Representations

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3 Project Overview

Automatic musical-instrument recognition is often described as a timbre-classification problem. In practice, however, instrument families do not occupy the same pitch ranges: bass notes are concentrated in lower registers, while other families may occur more often in middle or upper registers. A classifier can therefore obtain a favourable score by exploiting the association between pitch and family rather than learning a representation of instrument identity that remains stable across register. This creates a potential confound in both model training and evaluation.

This project investigates the following primary research question:

When classifying previously unseen instrument instances, how strongly does instrument-family recognition depend on pitch/register, and do pretrained music representations preserve instrument-family identity across pitch more effectively than conventional acoustic features?

The study uses the NSynth dataset and compares two representations under the same linear classification protocol: (1) a 76-dimensional handcrafted representation containing MFCC, spectral, energy, temporal, and attack descriptors; and (2) a frozen 768-dimensional embedding extracted from the final hidden layer of MERT-v0-public. A conventional test condition will first measure family recognition on the naturally occurring pitch distribution. The same fitted systems will then be evaluated without retraining on a fixed test manifest that balances family counts within register and velocity cells. A separate linear pitch probe will measure how much register information remains decodable from each representation. Finally, matched acoustic examples will be analysed to connect quantitative behaviour with spectral-envelope and onset characteristics.

The project is designed to distinguish three related but different questions: whether a representation contains pitch information, whether family recognition remains accurate after pitch distribution is controlled, and whether any register dependence is consistent across instrument families. Keeping these questions separate prevents high pitch decodability from being incorrectly interpreted as poor instrument recognition.

4 Background and Motivation

Timbre is a multidimensional perceptual property rather than a single measurable quantity. Classic perceptual studies identify spectral energy distribution and temporal attack behaviour as important dimensions of timbral similarity [1]. Later work likewise relates perceived brightness and instrument-sound dissimilarity to spectral centroid and attack time [2]. Comparative instrument-recognition research has also shown the value of cepstral, spectral, and temporal-development features [6]. These findings motivate the use of both spectral and temporal descriptors in conventional instrument-recognition systems.

Pitch and timbre are conceptually distinct, but they are not independent in recorded musical notes. Changing the fundamental frequency changes harmonic spacing, the interaction between harmonics and resonances, and potentially the excitation and radiation behaviour of the source. Consequently, a descriptor intended to represent timbre may also carry substantial pitch information. Dataset composition can amplify this interaction: if particular families appear mostly in particular registers, the label can be predicted partly from pitch distribution. More generally, benchmark scores can reflect dataset-specific regularities rather than only the intended concept [7]. A high ordinary test score would therefore not necessarily demonstrate pitch-robust recognition.

NSynth is well suited to a controlled study because it contains 305,979 four-second monophonic notes at 16 kHz, together with instrument-family, source-instrument, pitch, velocity, and source-type annotations [3]. Its official training, validation, and test splits separate source instruments, making it possible to evaluate recognition on instrument instances not observed during training. At the same time, NSynth is not naturally balanced across family and pitch, so its metadata must be audited before a fair register-controlled comparison is constructed.

Recent self-supervised music models offer an alternative to manually designed acoustic descriptors. MERT was pretrained for general music understanding using acoustic and musical teacher targets and has demonstrated transferable performance across multiple downstream tasks [4]. However, a strong pretrained representation can encode both desired and nuisance variables. Comparing absolute classification scores alone cannot establish whether MERT is more invariant to register. This project therefore combines controlled evaluation with a diagnostic pitch probe: family classification measures useful label information, while the probe measures linearly accessible register information in the same frozen representation.

5 Research Questions and Hypotheses

The project addresses four questions:

1. How much does instrument-family performance change when the test distribution is balanced across register and velocity?
2. Does a frozen pretrained music representation provide higher family-recognition performance, or a smaller ordinary-to-balanced performance change, than handcrafted acoustic features?
3. How accurately can lower, middle, and upper register be decoded from each representation using a linear probe?
4. Are register effects uniform, or do they differ across instrument families and acoustic conditions?

The working hypotheses are that the naturally distributed evaluation may reward pitch-family associations; that MERT will provide stronger absolute family recognition because it was pretrained on large-scale music audio; and that both representations will retain measurable pitch information. The comparison is deliberately capable of rejecting the stronger claim that a pretrained representation is more pitch-robust: this claim will be supported only if its family performance is consistently stable under the controlled evaluation, not merely because it achieves the highest ordinary score.

6 Proposed Methodology

6.1 Dataset Audit and Controlled Scope

The official NSynth train, validation, and test splits will be preserved. Metadata and audio-file coverage will first be verified, including unique item identifiers, valid annotations, expected audio duration and sampling rate, and absence of source-instrument overlap between the training and evaluation splits. All subset decisions will be made from metadata and fixed before classifier predictions are inspected.

A reproducible eligibility rule will select families and a shared pitch interval with adequate support. Each retained family must contain at least four distinct source instruments in both official evaluation splits. Candidate intervals must be divisible into three equal register bins, and every family–register–split cell must satisfy minimum note, instrument, and pitch-coverage requirements. Applying this rule defines the following controlled scope:

Design item	Fixed choice
Instrument families	bass, brass, guitar, keyboard, organ, string
Common pitch interval	MIDI 43–69
Register bins	lower 43–51; middle 52–60; upper 61–69
Random seed	5305
Primary balanced manifest	6,078 notes: 3,600 train, 1,884 validation, 594 test

Within each split of the primary manifest, every family will have the same number of notes in each register $\times$ velocity cell. At least two source instruments must contribute to each cell, and no one instrument may supply more than 50% of that cell. This design removes the most direct association between the family target and register or velocity while limiting domination by a single source instrument. A stricter supplementary manifest will match family counts within exact MIDI pitch $\times$ velocity cells; because this produces a much smaller test set, it will be treated as a sensitivity analysis rather than the main evaluation.

6.2 Audio Representations

All audio will be processed at 16 kHz. Two fixed-dimensional representations will be compared.

Handcrafted acoustic representation. For each note, frame-level descriptors will be extracted using a common short-time Fourier transform implemented with the `librosa` audio-analysis library [8]. They comprise 13 MFCCs and their first-order deltas; spectral centroid, bandwidth, roll-off, flatness, and flux; RMS energy and zero-crossing rate; and global energy-envelope descriptors including crest factor, temporal centroid, attack location, attack duration, and attack slope. Frame-level measurements will be summarised by means and standard deviations, with additional RMS quantiles and extrema, to form a 76-dimensional vector. These descriptors cover established spectral-envelope and temporal correlates of musical timbre while remaining interpretable [6].

Frozen pretrained representation. The same notes will be passed through `m-a-p/MERT-v0-public` at its required 16 kHz sample rate. Model weights will remain frozen. The final hidden layer will be mean-pooled over time to produce one 768-dimensional float32 embedding per note. The model revision, layer, pooling rule, and cache signature will be pinned so that the representation cannot change silently across runs.

6.3 Instrument-Family Classifiers

One classifier will be trained for each representation. A `StandardScaler` followed by a class-weighted linear support vector classifier will be fitted on official-training notes from the six selected families. The regularisation parameter C will be selected from $\{0.01, 0.1, 1.0\}$ using validation macro-F1 only. The test set will not be used for preprocessing, fitting, or model selection. A linear classifier is chosen so that differences are primarily attributable to the representation rather than to downstream model capacity [5].

The primary comparison will use the same fitted scaler and classifier in two test conditions:

1. **Ordinary test:** all eligible test notes from the six families in MIDI 43–69, retaining the natural family–pitch distribution.
2. **Pitch-balanced test:** the fixed 594-note test manifest with equal family counts within every register $\times$ velocity cell.

Neither system will be refitted between these conditions. The ordinary-to-balanced change is therefore interpretable because the representation, classifier, family set, and pitch interval remain constant; only the evaluation distribution is controlled. The balanced training and validation rows will be reserved for supplementary training-condition experiments rather than mixed into the primary comparison.

6.4 Pitch Probe and Register-Level Analysis

Separate linear probes will predict one of the three register bins from each frozen representation. A linear probe tests whether a target attribute is directly recoverable from a fixed representation without changing that representation [9]. Probe scalers and classifiers will be fitted only on official-training notes within MIDI 43–69, with C selected on the official validation split. Instrument-family labels will not be provided to the pitch probes. Evaluation on both the ordinary and balanced test sets will reveal whether pitch remains linearly accessible after family distribution is controlled.

Pitch-probe performance will be interpreted jointly with family recognition. High probe accuracy means that register information is encoded, but it does not by itself imply that family classification depends on that information. Evidence for dependence instead comes from a reduction in family performance after balancing, from systematic register-level differences, or from family-specific recall patterns.

6.5 Acoustic Error and Pair Analysis

Quantitative results will be complemented by metadata-controlled acoustic examples chosen without reference to classifier predictions. Same-pitch/different-family pairs will match MIDI pitch, velocity, and source type, while same-instrument/different-pitch pairs will hold source instrument and velocity fixed across distant registers. Waveforms, magnitude spectra, log-Mel spectrograms, Mel-smoothed spectral envelopes, estimated fundamental frequency, spectral centroid, and RMS attack envelopes will be compared. This analysis is explanatory rather than inferential: a small number of pairs can illustrate mechanisms but cannot estimate population-level effects.

6.6 Evaluation and Reproducibility

Accuracy and macro-averaged F1 will be reported for overall performance. Per-family recall, per-register macro-F1, family-by-register recall, and normalized confusion matrices will identify local failure modes that an aggregate score may hide. The main measure of possible confounding will be the change in family macro-F1 from the ordinary to the balanced test condition. All manifests, configurations, trained linear models, prediction tables, metrics, and figures will be saved in the repository. Fixed random seeds, resumable feature caches, validation-only model selection, and explicit one-to-one cache checks will support reproducibility.

7 Expected Outcomes and Contribution

The project is expected to deliver a controlled empirical account of pitch/register in NSynth instrument-family recognition, rather than only another leaderboard-style classifier comparison. Its main contributions will be:

- a documented metadata audit showing which families and pitch intervals support a defensible comparison;
- a frozen register- and velocity-balanced evaluation manifest with source-instrument contribution controls;
- a like-for-like comparison between interpretable handcrafted descriptors and a frozen pretrained music representation using the same linear protocol;
- a pitch-probe analysis that separates information content from task robustness; and
- an acoustic interpretation of representative within-pitch family differences and within-instrument register changes.

Several outcomes are possible. A large reduction after balancing would indicate that the ordinary score benefited from family-specific pitch distributions. A small change would suggest greater stability under the controlled distribution, although it would not establish pitch invariance. Similarly, higher MERT accuracy would support its usefulness as a transferable representation, but stronger pitch-probe performance would show that it still preserves pitch information. The most informative conclusion may therefore concern family- and register-specific behaviour rather than a single global ranking.

8 Limitations and Risk Management

NSynth contains isolated, fixed-duration notes rather than polyphonic musical performances, so conclusions will be limited to monophonic note-level recognition. Family labels are broad, and recordings include acoustic, electronic, and synthetic sources whose production chains may introduce cues unrelated to physical instrument construction. Balancing improves internal control but reduces test-set size, especially for exact-pitch sensitivity analysis; small differences will therefore be interpreted cautiously rather than presented as causal or statistically conclusive.

Computational cost is concentrated in MERT inference. Embeddings will be extracted once into validated, resumable shards, after which all classifiers and probes can run from cached arrays. The model revision and dependency versions will be pinned. If exact-pitch matching yields insufficient support for reliable family-by-register analysis, it will remain a supplementary robustness check and the larger register-balanced manifest will remain primary.

References

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